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Editorial

An annual review of key advances in evidence-based plastic surgery, A synopsis from the leads of the BAPRAS Research & Innovation, Education, and Trainees Committees

Understanding and practising evidence-based medicine is crucial in delivering optimal patient care in Plastic Surgery. It can be incredibly challenging for Plastic Surgeons with a demanding clinical workload to keep up to date with research and regulatory advances within the field. This is particularly difficult given the rate of development across such a vast surgical specialty. In this editorial, provided by the research and educational leads of PLASTA and BAPRAS, landmark articles and guidelines have been summarised to provide an overview of the key advances in Plastic Surgery over the past 12 months. This is not an exhaustive list but should act as a highly efficient resource for all Plastics Surgeons looking for an annual synopsis of clinical developments within their speciality. Topics have been divided into 8 subcategories: Breast Surgery, Burns, Cleft and Craniofacial, Cosmetic/Aesthetics, Hand, Head and Neck, Peripheral Nerve and Skin Cancer.

Breast surgery

There have been several highly impactful developments in Breast Surgery research over the past 12 months which offer a nuanced perspective on therapeutic decision-making.

In the pursuit of tailoring treatments for older women with hormone receptor-positive early breast cancer, Kunkler et al. conducted a phase 3 randomised trial assessing the omission of radiotherapy post breast-conserving surgery.[1] Their findings, spanning a 10-year follow-up as part of the PRIME II trial were published in NEJM. Omission of radiotherapy was associated with an increased incidence of local recurrence but had no detrimental effect on distant recurrence as the first event or overall survival among women 65 years of age or older with T1 or small T2 ER-Positive breast cancer.

Strnad et al.'s GEC-ESTRO trial delved into breast irradiation techniques, presenting 10-year results comparing accelerated partial breast irradiation (APBI) using multicatheter brachytherapy to whole-breast irradiation with a boost.[2] The study reinforced APBI using multicatheter brachytherapy after breast-conserving surgery in patients with early breast cancer is a valuable alternative to whole-breast irradiation in terms of treatment efficacy and is associated with fewer late side effects.

Shifting focus to axillary management, the AMAROS trial compared axillary lymph node dissection (ALND) and axillary radiotherapy (ART) in sentinel node-positive cT1-2 breast cancer.[3] The 10-year outcomes favoured ART, showcasing reduced lymphedema rates and comparable outcomes in overall survival, disease-free survival, and locoregional control. This highlights the importance of tailored axillary strategies in optimising outcomes.

In a ground-breaking phase 2 trial across seven centres, Kuerer et al. challenged conventional breast surgery norms.[4] Their study explored the possibility of eliminating the need for breast surgery in patients with early-stage triple-negative or HER2-positive breast cancer exhibiting a complete pathological response with neoadjuvant systemic therapy (NST). They evaluated radiotherapy alone, without breast surgery with NST who had an image-guided vacuum-assisted core biopsy (VACB) determined pathological

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complete response. It was shown that breast surgery may not be needed in patients with an image-guided VACB-determined pathological complete response following NST. While promising, the need for additional trials and longer-term endpoints remains imperative to validate the feasibility of this approach.

Mastectomy reconstruction options, crucially chosen on an individual basis, were scrutinised in a meta-analysis by Stefura et al., involving 55,455 patients across 32 studies.[5] This study compared the outcomes of autologous breast reconstruction (ABR) and implant-based breast reconstruction (IBR). Importantly, there was comparable safety outcomes between the two methods. ABR demonstrated significantly improved aesthetic and reconstructive satisfaction. However, it correlated with significantly higher costs (standard mean difference −0.69; $p = 0.010$).

Together, these diverse studies intricately contribute to refining the nuances of breast surgery decisions. They emphasise the paramount importance of personalised approaches and evidence-based practices in the ever-evolving landscape of breast cancer care.

Burns

There have been several innovative advances in burns surgery and management over the past 12 months. An exciting new app developed by Toolaroud et al. aimed to address the challenges faced by caregivers of children with severe burns by developing and evaluating a mobile-based self-management application called Burn.[6] Caregivers commonly identified post-discharge challenges related to infection control, wound care, and physical activity. The Burn application, designed with features like user registration, educational materials, caregiver-clinician communication, a chat box, and appointment booking, received positive usability scores, indicating its effectiveness in supporting caregivers and meeting specialists' and patients' needs. The co-design process with healthcare specialists and user evaluations contributed to the programme's utility and usability, making it a promising tool in paediatric burn care.

Meek micrografting was a technique first described in 1958 and involves the transplantation of multiple micrografts over a burn wound.[7] This provides 1:9 expansion of autologous grafts allowing coverage of a large TBSA burns with limited donor sites availability. Rijpma et al. provided a recent update on the outcomes for the Meek Micrograft technique from 15 articles comprising of 310 patients. Weighted averages reported graft take to be 82 ± 7%, time to wound closure 53 ± 20 days and length of hospital stay 61 ± 31 days. They concluded that overall study quality and data surrounding scar assessment was limited highlighting the need for future randomised controlled trials.

Several teams have been working on viable biological burns dressings that aid in the physiological processes of healing to improve aesthetic and functional outcomes. One such dressing is bacterial cellulose, which has been shown to significantly reduce scar formation and healing time. Despite these advantages bacterial cellulose dressings lacks anti-microbial properties. To address this Pasaribu et al. developed and tested a modified bacterial cellulose dressing, impregnated with chitosan and collagen.[8] They tested this compound for its tensile strength, anti-microbial properties, toxicity, and granulation to show this compound is stable and resulted in fewer bacteria on the dressing during in-vivo experiments. In-vivo test indicated that bacterial cellulose composite wound dressing supported the wound healing process on superficial partial burns.

A recent systematic review identified that burns patients ($n = 762$) consistently reported moderate levels of low self-esteem.[9] This highlights the importance of psychological support in these patients via an interdisciplinary approach. Aromatherapies have been proposed as an important non-pharmacological treatment adjunct for burns patients suffering from anxiety and pain. Two well-known aromatherapies, Lavender and Rosa damascene have been thought to help patients with these issues but have lacked higher-level evidence advocating for their use. Farzan et al. conducted a systematic review and meta-analysis of the impact of Rosa damascene and lavender on pain and anxiety.[10] They found Rosa damascene was associated significant reduction in pain and anxiety during dressing changes. Although aromatherapy with lavender decreased pain in the patients, it was not statistically significant.

Cleft and craniofacial

There have been several interesting developments in cleft and craniofacial surgery over the past year. Sakran et al. reported the development the Sommerlad-furlow (SF) modified cleft palate repair technique which combines aspects of both individual procedures (Sommerlad and Furlow) without a relaxing incision.[11] They report their outcomes from 320 patients treated with the technique split into < 1-year olds ($n = 143$), 1-2-year-olds ($n = 113$) and > 2 year olds ($n = 64$). 96.6% had complete wound healing with 81.56% having proper velopharyngeal function (VPF). Although velopharyngeal insufficiency was much greater in those > 2 years old at 45.4% but 9.8% in those < 1 year old. The SF technique demonstrates good post-operative outcomes in those < 2 years old, adding to other operative techniques to repair a cleft palate.

Correcting the cleft nasal deformity when carrying out unilateral cleft lip repair is likely to affect nasal and maxillary development later in life due to the septopremaxillary ligament transmitting a force that induces growth of the maxilla. Although surgeons have developed techniques since that have resulted in better outcomes of primary rhinoplasty during cleft lip repair, there is still a debate on whether it should be performed. Zelko et al. conducted a systematic review and meta-analysis to assess the evidence for and against primary unilateral cleft rhinoplasty.[12] 15 articles supported primary rhinoplasty during cleft repair with 5 of these studies highlighting that 43–100% of patients avoided revision rhinoplasty at 6 years follow up. There were significant differences in rhinoplasty technique leaving these findings closer to representing the outcomes of nasal intervention during cleft palate compared to doing nothing. With that in mind, there is a good evidence base to justify rhinoplasty during unilateral cleft lip repair.

In Craniofacial surgery, computer assisted magnetic navigation systems have used magnets for live image anatomy and aided in precise instrument control, reducing the incidence of surgical complications. While its use in precision craniofacial surgery can be significant, Sun et al. carried out an analysis of which surgical theatre instruments cause interference in the magnetic systems and to what magnitude.[13] They found that widely used stich scissors, teeth forceps and needle holders significantly impact the accuracy of the magnetic system prompting further thought for manufacturers to evaluate the electromagnetic interference of their equipment.

The gold standard in craniofacial bone reconstruction is autologous tissue however, it can be challenging to achieve accurate restoration of the skeletal contour. Through advances in tissue engineering, biosynthetic materials have become a viable alternative to autologous bone reconstruction. One such material is ceramic bone made from hydroxyapatite and b-tricalcium phosphate, which acts as a scaffold for osteoconduction and provides growth factors for osteoinduction. However ceramic bone cannot currently replace load-bearing bones owing to its fragility. Kim et al. tested a novel composite of ceramic bone and bioactive glass which gives the ceramic bone better structural rigidity, in a prospective trial.[14] 7 patients were enroled and were shown to have significant improvements in post-operative symmetry and only one patient had delayed wound healing due to previous radiation. It will be interesting to see the longer-term patient outcome data from Kim et al.

Cosmetic and aesthetic

Over the past 12 months, the field of aesthetic surgery has witnessed significant regulatory and research advancements, shaping the landscape of procedures and patient safety. In this review, we delve into key developments, ranging from updated guidelines on lipomodelling in aesthetic breast surgery to innovative approaches in pre-operative consultation using artificial intelligence.

This year BAPRAS, in collaboration with BAAPS and the ABS, has released an update to the 2012 publication on lipomodelling.[15] The guideline comments on the importance of accessibility of fat donor sites and the likelihood of including adipose-derived stem cells, which are particularly abundant in the lower abdomen. This inclusion enhances fat retention, vascularisation, and adipose regeneration. Notably, the guideline advocates for strategic use of local anaesthesia (LA) after fat harvest in general anaesthesia (GA) procedures, citing cytotoxic effects observed in cell culture experiments. To mitigate these effects in LA procedures, centrifugation and washing are recommended. Additionally, the guideline discusses experimental studies employing microneedling 1 week prior to grafting and platelet-rich plasma to enhance fat grafting outcomes.

Continuing the trajectory of safety improvement, BAAPS provided updated guidelines on fat grafting in superficial gluteal lipofilling.[16] They recalibrated the risk of mortality from fat embolism in Brazilian Butt Lift, which was previously reported to be 1:2500. The actual risk was later estimated to be nearer 1:15,000, which is comparable to the mortality risk in abdominoplasty. This improvement in safety is principally due to intraoperative ultrasound guidance, new technologies, surgical techniques, and training being developed to avoid intramuscular injection of fat. The risk of serious complications is considerably higher in unregulated facilities or via medial tourism.

Further developments in gluteal aesthetic surgery questioned the long-term aesthetic and functional outcomes of intramuscular augmentation gluteoplasty and its effect on gluteus maximus muscle atrophy.[17] CT 3D volumetric analysis demonstrated median atrophy of 6.68% of gluteus maximus muscle volume in 12 months but reported no correlation between volume of implant and atrophy percentage of gluteus maximus muscle. Importantly, they also reported recovery in muscle volumetry in patients who practiced physical activity, which should be considered in pre-operative consultation.

Xie et al. proposed an advancement in pre-operative consultation by leveraging ChatGPT, an artificial intelligence language model to provide valuable advice and counselling for rhinoplasty consultations.[18] This innovative approach is particularly beneficial for hesitant patients or those with limited access to medical advice. However, they did report that further research is needed to determine the scope and limitations of AI language models in this domain and to assess the potential benefits and risks associated with their use.

Beyond procedural innovations, recent months have witnessed legislative changes, most notably the implementation of the Botulinum Toxin and Cosmetic Fillers (Children) Act.[19] This legislation marks a significant ethical stride by criminalising the administration of botulinum toxin or fillers for cosmetic purposes to individuals under 18 in England. While maintaining strict boundaries, the law allows registered doctors to approve treatments for individuals under 18 if deemed clinically necessary, reflecting a balanced approach to ethical practice.

These diverse developments collectively underline the dynamic nature of aesthetic surgery, encompassing evolving guidelines, technological strides, and legislative changes. As the field propels forward, continued research and collaboration remain imperative to ensure the highest standards of safety, efficacy, and ethical conduct in aesthetic surgery.

Hand

Recent shifts in hand surgery management and services have been the focus of investigation. The RSTN's investigation brought to light the transformative effects of the COVID-19 pandemic on hand trauma management in the UK and Europe.[20] The study, conducted across 35 hand surgery units, encompassed 2540 patients. Notably, there was a 3-7% increase in non-operative management, except for flexor tendon injuries. Consultant triage doubled, with a 22% rise in the see-and-treat model. The adaptation to low-resource settings saw a 13% increase in minor operating theatres and a 10% rise in clinic rooms. The use of WALANT, absorbable sutures, and remote follow-ups increased by 16%, 24%, and 11-25%, respectively. Remarkably, there was no increase in the complication rate, suggesting a potential shift towards a modern, evidence-based, and environmentally sustainable approach to hand trauma services.

Addressing long-term shifts in hand trauma management, a consensus paper by six eminent hand surgeons delves into the changes in flexor tendon repair over the past two decades.[21] The report highlighted a recent trend towards robust multi-stranded core suture techniques with simplified epitendinous sutures. Although, they do not conclude which multi-strand core suture is best. Venting of the critical pulleys over less than 2 cm length was considered safe and favoured functional recovery. These repairs and recent motion protocols lead to remarkably more reliable repairs, with over 80% good or excellent outcomes achieved rather consistently after Zone 2 repair along with infrequent need for tenolysis. They reported outcomes of Zone 2 repairs not dissimilar to those in other zones with minimal rupture incidence.

Transitioning to thumb injuries, the BSSH's evidence-based guidelines for acute ulnar collateral ligament injuries provide a structured approach.[22] They have provided several key clinical practice recommendations. Recommendations include clinical examination for UCL laxity, orthogonal X-rays for fracture assessment, and a nuanced stance on the use of ultrasound or magnetic resonance imaging. Patients without significant joint laxity (no firm endpoint, > 20-degree laxity vs other side, > 30-degree laxity), should be treated non-surgically with rigid orthosis. They report that it is reasonable to offer early surgery or non-surgical immobilisation of the MCPJ to patients with significant joint laxity on clinical examination.

Advances have also been reported in elective hand surgery. An article in PRS investigated whether web-based patient-reported outcome measures (PROMs) could help surgeons remotely assess the need for examination and subsequent treatment of patients with Dupuytren disease (DD).[23] The results showed that PROMS measured around 1 year before treatment can predict treatment for DD. Paving the way for telemonitoring and optimising patient care and treatment effectiveness in DD.

Expanding the horizon, innovations in hand rejuvenation techniques offer a holistic approach to hand surgery. Rohrich et al.'s description of a single-access incision combined with the massage technique and fat grafting showcases a technique that not only enhances skin texture and appearance but also addresses visible superficial veins.[24] This method, recommended as the standard of care for moderately to severely aged hands, exemplifies the continuous evolution in hand surgery.

Collectively, these developments underscore the dynamic nature of hand surgery, where adaptation, innovation, and a patient-centric approach converge to shape the future of this intricate field.

Head and neck

The landscape of head and neck cancer care has undergone significant shifts, notably marked by updates in the NCCN Head and Neck Cancer guidelines.[25] These changes reflect the increasing incidence of HPV-positive oropharyngeal cancer, contrasting with a decline in HPV-negative cases primarily linked to tobacco and alcohol use. As we navigate through these updates, it becomes clear that a deeper understanding of the distinct characteristics and responses to treatment in HPV-positive and HPV-negative tumours is essential. The rising incidence of HPV-positive oropharyngeal cancer, estimated at 60-70% in the US and parts of Europe, has prompted a re-evaluation of treatment strategies. Locally advanced HPV-positive tumours exhibit improved responses to treatment, leading to enhanced overall survival (OS) and progression-free survival (PFS) compared to HPV-negative tumours. The NCCN Guidelines now recommend tumour HPV testing for all oropharyngeal cancer patients, paving the way for research into treatment de-intensification, such as dose-reduced radiation therapy (RT). While promising, the implementation of RT de-intensification for HPV-positive cancers awaits conclusive data from randomised phase III trials, with considerations for excluding patients with T4 or N3 disease.

Concerns surrounding radiotherapy de-intensification persist due to potential radiation-induced toxic effects, such as osteoradionecrosis of the jaw in patients with head and neck cancer. A comparative study from Memorial Sloan Kettering Cancer Center evaluating proton radiation therapy versus conventional photon radiation therapy sheds light on this issue.[26] Despite the advantages of proton therapy in reducing mucositis and xerostomia, the prevalence of osteoradionecrosis following proton radiation therapy remains a clinical challenge affecting 10.6% of the 122 patients in the study. This emphasises the need for ongoing scrutiny of de-intensification strategies.

The impact of smoking status on treatment de-intensification takes centre stage, as evidenced by Ma et al.'s cohort study encompassing 518 patients with head and neck cancer.[27] The study identifies a threshold of 22 pack-years of smoking associated with survival and distant metastasis outcomes. Notably, current smoking status is linked to adverse outcomes specifically in patients with HPV-associated head and neck cancer. These findings underscore the intricate interplay between smoking, treatment response, and the potential for de-intensification.

Moving beyond clinicopathological features, biomarkers have emerged as valuable tools for risk stratification in head and neck cancer. Cell-free nuclear DNA and cell-free mitochondrial DNA have shown promising diagnostic utilities among patients with head and neck squamous cell carcinoma. This prognostic tool, known as liquid biopsy, was acquired from saliva-based cell-free and mitochondrial DNA in 94 patients, avoiding the need for additional tissue biopsy or blood tests.[28] Univariant analysis showed that only the absolute copy number of cf-mtDNA was the sole predictor of overall survival. This confirms that saliva is a reliable and non-invasive source of information that can be used to predict the overall survival of patients with head and neck squamous cell carcinoma.

In the dynamic landscape of head and neck cancer, the recent updates in guidelines, considerations for deintensification, and the exploration of biomarkers underscore the importance of tailored approaches for different subtypes. As we delve into the intricacies of treatment strategies, ongoing research and a nuanced understanding of patient characteristics will continue to shape the evolution of head and neck cancer care.

Peripheral nerve

Peripheral nerve injuries (PNI) are associated with reduced motor and poor sensory post-operative outcomes. Aman et al. investigated the epidemiology and aetiology of PNI in one of the largest cohorts to date.[29] 5026 patients were included and comprised predominantly of young males suffering from non-work related PNI with the main cause being lacerations (54.6%) to the upper limb (88.3%). PNI can be masked by the larger injury it is within such as a fracture, this is particularly prevalent in proximal-level injuries which are more likely to be associated with fractures. Distal nerve injuries were more likely to be associated with vessel or tendon injury. Early return to work and optimising long-term functional outcomes are facilitated by the relatively young age of affected patients, prompt diagnosis, and treatment at specialised centers. The data emphasise the significance of dedicated attention to nerve injuries, which might be overshadowed by more extensive accompanying injuries.

The emphasis on the accurate and early diagnosis of PNI has been further explored.[30] Follow-up for these patients often comprises of clinical examination to assess patient-reported outcome measure and sometimes electroneurography - however this test can cause delays in treatment and potentially impaired healing. MR Neurography (MRN) is being pioneered and can visualise nerve healing. Aman et al. proposed to investigate how well MRN correlates with the clinical progression of reinnervation. It was shown that MRN can visualise nerve injuries on a fascicular level, providing not only early diagnosis and therapy decisions but also providing a precise tool for monitoring reinnervation processes.

The post-operative outcomes of PNI can also worsen with age due to poorer nerve regeneration because of cellular homoeostasis imbalance associated with ageing. Although the exact mechanisms of this have not been fully elucidated. Maita et al. reports through a systematic review that this was caused by chronic inflammatory state which delays macrophages response to injury and altered gene expression patterns in Schwann cells leading to them becoming dysfunctional.[31] This highlights that further research should focus on these cells and their interaction to develop therapies for nerve regeneration. One such nerve regeneration therapy already in use is a vein conduit which helps guide axonal growth towards the distal target, like a tunnel and prevents unwanted sprouting. Substances such as platelet-rich fibrin (PRF) have been added to conduits as they contain growth factors to help create a better cellular microenvironment to theoretically improve regeneration. Atti et al. investigated the functional outcomes after repairing the sciatic nerve in rats with PRF and vein conduits. They found nerve grafts with vein conduits to exhibit the best functional outcome at 90 days with no added benefit when prefilled with PRF.

Nerve regeneration is also important in the sensorimotor control of prosthetics. Rat studies have using dermal sensory regenerative peripheral nerve interface as a biological interface to provide high-fidelity sensory feedback in prosthetics.[32] This was done by securing fascicles of residual sensory peripheral nerves into autologous dermal grafts. Findings suggested afferent nerve fibres successfully reinnervate dermal regenerative peripheral nerve interfaces evoking a graded stimulus response like normal skin.

We have witnessed significant strides in the realm of peripheral nerve interfacing and regeneration, signifying noteworthy progress in comprehending recognised injuries.

Skin cancer

In the melanoma field, momentum is building behind an evolving understanding of the benefits of neoadjuvant treatment. In this update, we delve into an emerging strategy for neo-adjuvant systemic therapy (NAST) and index lymph node resection (ILN; the largest lymph node metastasis at baseline imaging) for high-risk macroscopic stage III melanoma. As the evidence base continues to shine a spotlight on the superiority of NAST over upfront surgery and adjuvant systemic therapies, these novel approaches look set to become the new standards of care for resectable metastatic melanoma.

Since the approval of BRAF-targeted therapy (TT) and immune-checkpoint blockade (ICB) based therapies for stage III and stage IIB/C melanoma in the adjuvant setting, patients with high-risk melanoma have been able to benefit from multiple systemic treatment options post-surgery. Research since has focussed on refining both the delivery of systemic treatment and the morbidity of lymph node surgery. Reijers et al. demonstrated the representative prognostic value of an initial major pathological response (MPR) to NAST within the ILN following therapeutic lymph node dissection (TLND).[33] These findings, and the substantial morbidity associated with TLND, raised the question of whether TLNDs could be safely omitted in patients exhibiting MPRs to NAST. It is the PRADO trial (Personalised Response-directed surgery and Adjuvant therapy after neoadjuvant ipilimumab and nivolumab for stage III melanoma) that perhaps offers the most tantalising glimpse into the future with a refined, personalised, response-driven approach to systemic treatment and lymph node surgery for resectable metastatic melanoma.

PRADO, a multicentre phase 2 trial, was designed to investigate the role of MPRs within ILNs only - as a means of determining subsequent TLND and adjuvant therapy. In this study, ILNs were marked under ultrasound guidance using a variety of potential markers (including magnetic Memaloc markers, nitinol UltraCor Twirl markers, hydrogel markers and radioactive I-125 seeds). After 6-weeks of NAST, an initial procedure was carried out to resect only the marked ILN. If a MPR was identified in ILNs (≤10% viable tumour) TLND was omitted, and that patient proceeded to clinical/radiological surveillance only. If a pathological partial response (pPR = 10-50% viable tumour) was identified, patients underwent TLND followed by clinical/radiological surveillance. If a pathological non-response (pNR ≥50% viable tumour) was identified, patients received both TLND and adjuvant systemic therapy. Ultimately, a total of 90 patient underwent ILN procedures in the PRADO study, of which 60 achieved MPR and 59 had TLND omitted as a result. Using this approach, the 24-month relapse-free survival and distant metastasis-free survival rates were 93% and 98% for MPR, 64% and 64% in patients with pPR, and 71% and 76% in patients with pNR. These results demonstrate the potentially strong benefits of NAST followed by a personalised surgical approach stage III metastatic melanoma.

Building on neoadjuvant approaches, Patel et al. explored the role of neoadjuvant pembrolizumab in resectable stage III or IV melanoma.[34] Their investigation compared the event-free survival of patients receiving neoadjuvant pembrolizumab before and after surgery versus those receiving adjuvant pembrolizumab alone after surgery. Significantly longer event-free survival was observed in the neoadjuvant group. Importantly, the study identified no new toxic effects, highlighting the potential safety and efficacy of this approach in the advanced melanoma setting.

In addition to advance melanoma, research has progressed in early-stage melanoma. Progress in early-stage melanoma research has focused on refining clinical practices, particularly regarding sentinel lymph node biopsy (SLNB) in T1a cutaneous melanoma. Currently, Sentinel lymph node biopsy is not routinely recommended for T1a cutaneous melanoma due to the overall low risk of positivity. However, recent findings indicate that T1a melanomas, with additional factors such as younger age, lymphovascular invasion, mitogenicity, and head/neck primary site posing a higher risk of SLN+.[35] This nuanced understanding contributes to a more tailored approach in early-stage melanoma management.

The evolving landscape of melanoma research has been reflected in recent updates in the NICE Melanoma Guidelines.[36] These guidelines aim to reduce the variation in clinical practice across the UK; including disparity with regard to the use of dermoscopy, photography, access to SLNB, and follow-up imaging. The key updates from the 2015 guideline are: BRAF testing is now being considered at earlier (Stages IIA and IIB), SLNB will not be offered to all melanoma patients with a Breslow thickness of 0.8-1.0 mm, there is a significant increase in the demand on ultrasound scanning as this is recommended for the draining nodal basin when SLNB was not done in stage IB-III disease and added to follow-up regimes, change in indication and frequency of CT scan follow-up, and use of magnetic resonance imaging instead of CT in pregnant or young women to reduce radiation burden.[36] These updates reflect a commitment to aligning clinical practices with the latest evidence and optimising patient care.

Research developments have also been witnessed in non-melanoma skin cancer. Cutaneous squamous cell carcinoma (cSSC) is the most common metastasising tumour in the world. Electrochemotherapy (ECT) is an alternative treatment option for unresectable locally advanced or recurrent cSSC. Bertino et al. analysed the International Network for Sharing Practice on ECT (InspECT) registry to evaluate the use of ECT in cSSC.[37] Analysis consisted of 162 patients from 18 European Centres. The response to treatment per patient was 62% complete and 21% partial, demonstrating objective antitumoural activity. In the multivariant model, intravenous bleomycin administration and tumour < 3 cm in diameter showed a significant association with a positive outcome from ECT.

The recent strides in skin cancer research, as highlighted by the PRADO trial, updates in NICE Melanoma Guidelines, and advances in ECT underscore the importance of personalised and evidence-based approaches. As the field continues to evolve, these advancements pave the way for enhanced treatment strategies and standardised clinical practices, ultimately improving outcomes for individuals affected by skin cancer.

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Declaration of Competing Interest

TWA - PLASTA Research Rep, RPJ - Research Chair BAPRAS, AF - Education Chair BAPRAS. All remaining authors have declared no conflicts of interest.

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